

A compact proper subset of a seminorm ball has smaller Haar measure

MathlibAnnex.SeminormBall.measure_lt

theorem

Finds a nonempty open part of the missing set and uses finiteness of the compact set to make the measure comparison strict.

Statement

Let E be a real normed space with its Borel measurable structure, let μ be an additive Haar measure, and let $p : E \rightarrow \mathbb{R}$ be a continuous seminorm. Put $B_p = \{x : p(x) \leq 1\}$. If $K \subsetneq B_p$ is compact, then

$$\mu(K) < \mu(B_p).$$

Assumptions

The measure is an additive Haar measure: in particular it is finite on compact sets and positive on nonempty open sets. The seminorm p is continuous for the given norm topology; it need not be definite. The set K is compact, is contained in B_p , and is not equal to B_p . No convexity of K , finite-dimensionality of E , or finiteness of $\mu(B_p)$ is added.

Conclusion

The measure inequality is strict in the extended nonnegative reals $[0, \infty]$. Compactness ensures $\mu(K) < \infty$, while $\mu(B_p)$ is allowed to be infinite.

The key geometric input is a nonempty interior of $B_p \setminus K$. It concerns the ambient norm topology, even when p has a kernel. Taking $p = \|\cdot\|$ gives the separately cited norm-ball specialization, without imposing convexity on K .

Proof route

Choose $y \in B_p \setminus K$. Compactness makes K closed, so its complement contains a neighbourhood of y . An inward point ty with $0 < t < 1$ sufficiently close to 1 remains outside K . By seminorm homogeneity,

$$p(ty) = tp(y) \leq t < 1.$$

Continuity of p therefore makes $\{z : p(z) < 1\} \cap K^c$ a nonempty open subset of $B_p \setminus K$. Positivity of Haar measure on nonempty open sets implies $0 < \mu(B_p \setminus K)$. The sets K and $B_p \setminus K$ are measurable and disjoint, with union B_p . Hence

$$\mu(B_p) = \mu(K) + \mu(B_p \setminus K) > \mu(K),$$

where strictness uses $\mu(K) < \infty$, obtained from compactness. This reasoning remains valid if the second summand is infinite.

▼ Proof steps

1. To make the inward choice explicit, take $\delta > 0$ with the open norm ball of radius δ about y contained in K^c .
Set

$$a = \min \left\{ \frac{1}{2}, \frac{\delta}{2(\|y\| + 1)} \right\}, \quad t = 1 - a.$$

Then $0 < a \leq 1/2$, so $0 < t < 1$, and

$$\|ty - y\| = a\|y\| < \frac{\delta}{2} < \delta.$$

Thus $ty \notin K$. The strict inequality for $p(ty)$ supplies an interior point of B_p ; intersecting its open neighbourhood with K^c gives the nonempty open defect.

2. The compact set K is closed and Borel measurable. Continuity of p makes $B_p = \{p \leq 1\}$ closed and hence measurable, so its difference with K is measurable. The containment $K \subseteq B_p$ gives the disjoint decomposition $B_p = K \cup (B_p \setminus K)$. Additivity and the finite compact measure now justify the strict inequality, without subtracting infinite measures.

Main citations

- Exact declaration and its source — `MathlibAnnex.SeminormBall.measure_lt`
- An open defect in a compact proper subset —
`MathlibAnnex.SeminormBall.interior_sdiff_nonempty`
- The explicit inward radial choice — `MathlibAnnex.SeminormBall.exists_radial_contraction`
- The norm-ball specialization — `MathlibAnnex.NormBall.measure_lt`

Lean source signature (exact)

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theorem measure_lt
  [NormedAddCommGroup E] [NormedSpace ℝ E]
  [MeasurableSpace E] [BorelSpace E]
  (μ : Measure E) [Measure.IsAddHaarMeasure μ]
  (p : Seminorm ℝ E) (hp : Continuous p)
  {K : Set E} (hKcompact : IsCompact K)
  (hsub : K ⊆ p.closedBall 0 1) (hne : K ≠ p.closedBall 0 1) :
  μ K < μ (p.closedBall 0 1)
```

In the source	Mathematical meaning
<code>[NormedAddCommGroup E] [NormedSpace ℝ E] [MeasurableSpace E] [BorelSpace E]</code>	The real normed space E carries its Borel measurable structure.
<code>(μ : Measure E) [Measure.IsAddHaarMeasure μ]</code>	The additive Haar measure μ , finite on compact sets and positive on nonempty open sets.
<code>(p : Seminorm ℝ E) (hp : Continuous p)</code>	The seminorm $p : E \rightarrow \mathbb{R}$ is continuous for the ambient norm topology; it may have a nontrivial kernel.
<code>p.closedBall 0 1</code>	The closed seminorm unit ball $B_p = \{x : p(x) \leq 1\}$.
<code>{K : Set E} (hKcompact : IsCompact K)</code>	The set $K \subseteq E$ is compact. This ensures $\mu(K) < \infty$; it does not assert compactness of B_p .

In the source	Mathematical meaning
<code>(hsub : K ⊆ p.closedBall 0 1) (hne : K ≠ p.closedBall 0 1)</code>	The containment and inequality together say $K \subsetneq B_p$.
<code>μ K < μ (p.closedBall 0 1)</code>	The conclusion $\mu(K) < \mu(B_p)$ in $[0, \infty]$. The right-hand measure may be infinite.

▼ Exact surrounding binder context (separate excerpt)

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noncomputable section

open Set MeasureTheory
open scoped ENNReal

namespace MathlibAnnex

namespace SeminormBall

variable {E F : Type*}

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▼ Exact content identity

Declaration: `MathlibAnnex.SeminormBall.measure_lt`

Accepted content SHA-256: `cf91ef813d756e6a74e3bad4ef03ea2d2e828b67ce7a7e9ab4ff1ebd54f8c6c0`

Accepted source guide SHA-256: `c99001ecaddc5c97f3a2895ac44b059d204ed125a1e2c6ff7e5724147ae1b72a`

Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`